

The Black Rapids Glacier

An unusual glacier, the largest in the Alaska Range, with a history of phases of exceptionally rapid advance



The Black Rapids Glacier flows down a valley in south central Alaska and is one of the world's more accessible glaciers, because its snout lies only a few miles from a road. It is about 25 miles (40 km) long and has several tributaries. In 2002, an earthquake caused a huge part of it to be buried in rock debris.

The Black Rapids is a surge-type glacier, in which periodic rapid transfers of ice occur from the upper to the lower parts, which increase in thickness and then

surge forward. The Black Rapids Glacier made national news in the U.S. for three months in 1937, when its ice suddenly began moving at 100 ft (30 m) a day—more than 20 times its normal rate—and its snout began to rapidly advance down its valley. During this period, it became known as the Galloping Glacier. Since this surge, however, the Black Rapids has steadily retreated to its current length, which is at a historic minimum.

▽ RARE FIN

An ice fin—caused by pressure squeezing ice upward as a glacier grinds against hard bedrock—is a rare glacial feature. This fin was sighted on the Black Rapids in 2013.



◁ SURFACE MELT

Surface pools, and even streams, can develop on the higher parts of the Black Rapids when meltwater forms faster than it can be absorbed by the glacier.

